

Question number	Answer	Additional guidance	Mark
7(a)	An explanation that includes four of the following points: • without chloroplasts the plant will not be able to {photosynthesise / <u>absorb</u> light} (1) • {bacteria living in the xylem / sticky gum} could prevent {water / mineral ions} {reaching the leaves / being transported (up)} (1) • no water for {photolysis / hydration / turgidity} (1) • no magnesium ions for chlorophyll (1) • no {GALP / glucose} will be synthesised (1)	NB ACCEPT less for no, throughout ACCEPT white parts / streaks will not photosynthesise no light-dependent reaction IGNORE mineral ions ACCEPT other named ion and function that would prevent photosynthesis from occurring or plant dying ACCEPT named molecule e.g. sucrose / chlorophyll biomass no ATP	(4)

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7(b)(i)	A description that includes the following points: - DNA in prokaryotic cells is {circular / looped} and in human cells is linear (1) - DNA in prokaryotic cells has (one) more phosphodiester bond than human cells (of the same number of nucleotides) (1)	PIECE TOGETHER THROUGHOUT IGNORE numbers of strands / double helix / plasmids / non-structural comments ACCEPT prokaryotes do not have 3' and 5' ends but human cells do ACCEPT prokaryotes have no unbound phosphate group but humans do	(2)

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7(b)(ii)	An answer that includes the following points: - because the bacteria will {die / not be able to grow / will not be able to divide} (due to the inability of DNA to supercoil) (1) - human cells do not have {DNA gyrase / the enzyme} / <u>only</u> bacteria have the enzyme affected (1) - therefore human cells will not be {affected / damaged / harmed} (1)	NB ACCEPT pathogen or bacteria or prokaryotic cell throughout ACCEPT bactericidal / bacteriostatic IGNORE consequences of no supercoiling DO NOT ACCEPT in context of no {chloroplasts / xylem}	(3)

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7(b)(iii)	An explanation that includes the following points: - because bacteria are becoming (increasingly) resistant to antibiotics (1) - so we will not be able to treat {antibiotic-resistant bacteria / diseases / infections} (1)	ACCEPT pathogens because bacteria will not be resistant to new antibiotics to win the evolutionary race DO NOT ACCEPT viruses ACCEPT so people will not die from bacterial infections	(2)

